HILBERT SCHEME

TYSON KLINGNER

1. Introduction

To understand the geometry of a given algebraic variety X we often need to understand some of its subschemes. For example, if $X = \mathbb{P}^n$, then all of its smooth degree d hypersurfaces are diffeomorphic. When studying subschemes one quickly sees that subschemes comes in families, e.g.,

$$V(xy - tz^2) \subseteq \mathbb{P}^2_{x,y,z} \times \mathbb{A}^1_t$$

defines a family of conics in $\mathbb{P}^2$. Hence, to understand subschemes we will study them in families. We achieve this via the *Hilbert functor*. Once we introduce the Hilbert functor we will show that the Hilbert functor is represetable by a projective scheme, which we call the *Hilbert scheme*. In what proceeds we work over $k = \overline{k}$. Every construction extends to the relative setting, but we work over k for simplicity.

2. Hilbert Functor

Let X/k be a projective variety and let $\mathcal{O}_X(1)$ be a fixed very ample line bundle. Before defining the Hilbert functor we will introduce the *Hibert polynomial.*

Definition-Proposition 2.1. Let $\mathcal{F}$ be a coherent sheaf on X, then there is a numerical polynomial $P(z) \in \mathbb{Q}[z]$, such that $\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for every $n \in \mathbb{Z}$. We call $P_{\mathcal{F}}$ the *Hilbert polynomial* of $\mathcal{F}$ with respect to $\mathcal{O}_X(1)$. If $Z \subset X$ is a closed subscheme, then we define the *Hilbert polynomial* of Z to be $P_{\mathcal{O}_Z}$.

Proposition 2.2 (Properties of Hilbert Polynomials)**.** *Here we collect some basics properties of Hilbert polynomials*

(1) *Suppose there is a short exact sequence of coherent sheaves*

$$0 \to \mathcal{E} \to \mathcal{F} \to \mathcal{G} \to 0.$$

Then,

$$P_{\mathcal{F}}(n) = P_{\mathcal{E}}(n) + P_{\mathcal{G}}(n)$$

(2)

Date: April 7, 2026.

Example 2.3. Suppose $X = \mathbb{P}^m$. Then, $\chi(\mathcal{O}_{\mathbb{P}^m}(n)) = \binom{m+n}{m}$ for $n >> 0$ so the Hilbert polynomial of $\mathbb{P}^m$ is given by

$$P_{\mathbb{P}^m}(n) = \binom{m+n}{m}, \quad n >> 0$$

In particular,

$$P_{\mathbb{P}^2}(n) = \frac{(n+1)(n+2)}{2}, \quad n >> 0.$$

If $C \subseteq \mathbb{P}^2$ is a degree d curve, then there is a short exact sequence

$$0 \to \mathcal{O}_{\mathbb{P}^2}(n-d) \to \mathcal{O}_{\mathbb{P}^2}(n) \to \mathcal{O}_C(n) \to 0$$

and since $P_C(n) = P_{\mathbb{P}^2}(n) - P_{\mathbb{P}^2}(n-d)$, it follows that

$$P_C(n) = dn - \frac{1}{2}d(d-3), \quad n >> 0.$$